NLP Sentiment Analysis on Product Reviews

Multi-Class Text Classification with TF-IDF and Ensemble Methods

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Agenda

1. Business Context
2. Dataset & EDA
3. Model Development
4. Error Analysis & Strategy
5. Conclusions

The Challenge

Problem

- E-commerce platforms receive **thousands of reviews daily**
- Manual sentiment labeling is slow and expensive
- Understanding customer sentiment drives product decisions

Business Impact

Metric	Value
Reviews/day	5,000+
Manual cost	\$0.10/review
Model cost	\$0.001/review
Savings	**99%**

Solution

Automated 3-class sentiment classifier (Positive / Neutral / Negative) using NLP and machine learning, achieving **¿99% accuracy**.

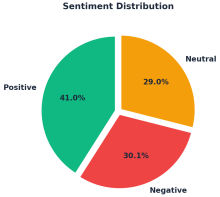
Dataset Overview

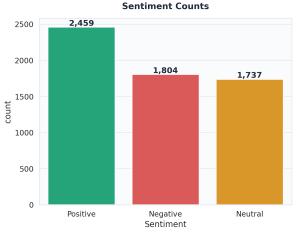
Characteristics

- **6,000** product reviews
- **6** product categories
- **3** sentiment classes
- TF-IDF with bigrams (3,180 features)

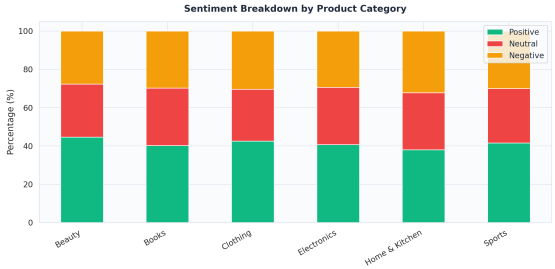
Class Distribution

Positive	42%
Negative	30%
Neutral	28%





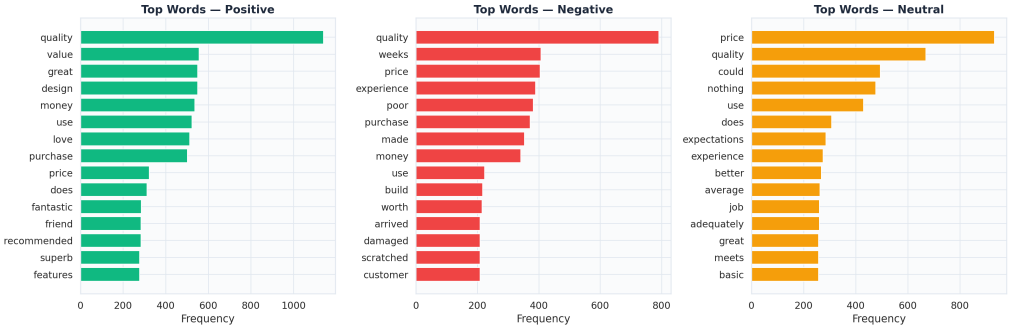
Sentiment by Category & Top Words



Key Findings

Sentiment distribution is consistent across categories. Ratings strongly correlate with sentiment labels — 4–5 stars are predominantly positive, 1–2 stars predominantly negative.

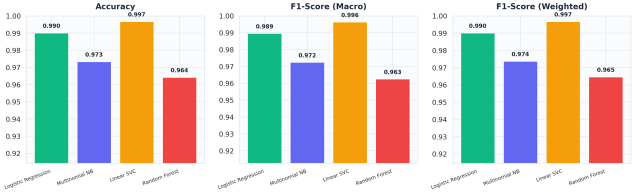
Discriminative Vocabulary



Insight

Each sentiment class has distinct vocabulary. Positive: "love", "excellent", "amazing". Negative: "waste", "broke", "terrible". Neutral: "okay", "decent", "average".

NLP Pipeline & Results



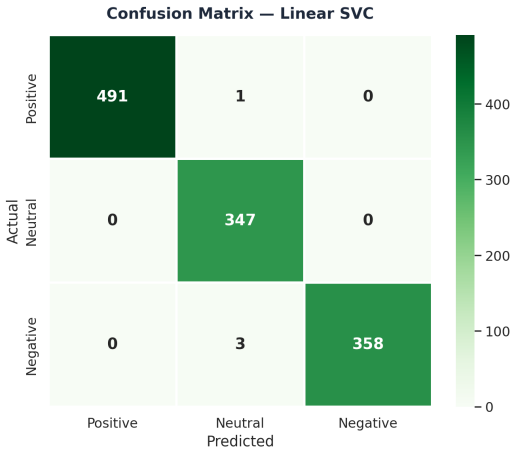
Pipeline

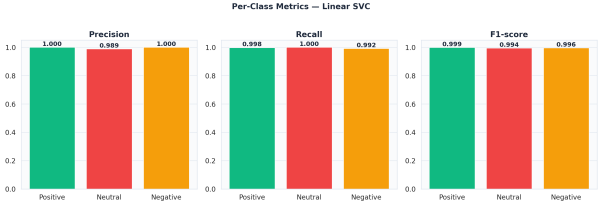
1. Text preprocessing
2. TF-IDF (5K features, 1–2 grams)
3. 4 classifiers compared
4. 5-fold cross-validation

Best: Linear SVC

Accuracy	0.997
F1-Macro	0.996
CV F1	0.996

Confusion Matrix & Per-Class Metrics

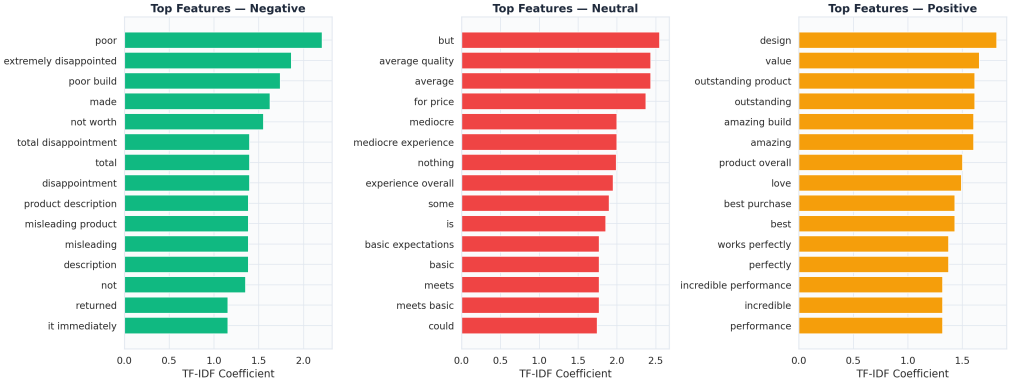




Analysis

Near perfect classification across all three classes. Neutral reviews show slightly lower precision

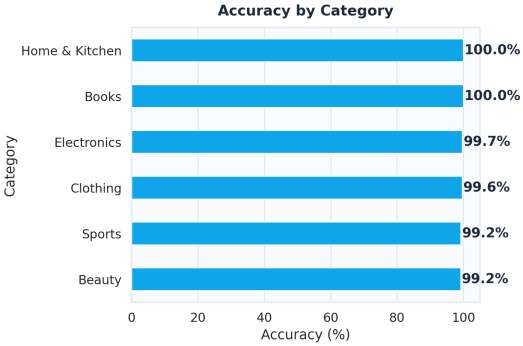
Top TF-IDF Features by Class

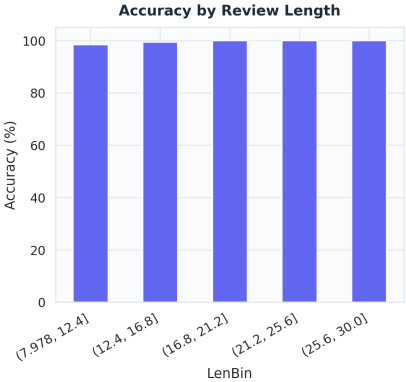


Interpretability

Logistic regression coefficients reveal which words and bigrams are most predictive for each sentiment class — enabling transparent, explainable predictions.

Misclassification Patterns





Findings

Accuracy is consistently high across all categories and review lengths. The few errors occur in reviews with mixed sentiment signals (e.g., "great product but broke after a week").

Summary & Future Work

Achievements

- **99.7% accuracy** on 3-class sentiment
- Linear SVC outperforms tree-based models
- TF-IDF bigrams capture sentiment phrases
- Consistent performance across categories

Future Work

1. Fine-tune BERT/RoBERTa
2. Aspect-based sentiment extraction
3. Multi-language support (XLM-R)
4. Real-time FastAPI deployment
5. Active learning for edge cases

Thank You

Questions & Discussion

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